Final Analyzed Data: `Project2-Data.csv` (cleaned & merged dataset)
Raw Data Source: `Project2-RawData.csv`
Unit of Observation: Each row represents a monthly observation (Year–Month) spanning 1980–2025.

1. Date (Month-Year)

- **Source & Processing:**
  Provided in the raw dataset and used as the key to merge CPI and Consumer Sentiment data. Converted into a consistent datetime format.
- **Observations: ~554**
- **Data Type: Time index**
- **Use in Analysis:**
  Enables time-series modeling and alignment across datasets

2. Consumer Sentiment Index (UMCSENT)

- **Source & Processing:**
  Collected from FRED (University of Michigan Survey of Consumers). Merged with CPI data by date.
- **Observations: ~554 (1 removed in cleaned dataset if applicable)**
- **Data Type: Quantitative (continuous)**
- **Role: Dependent variable in all models (OLS, Lasso, ARIMAX)**
- **Description: Measures consumer optimism/pessimism about the economy**

3. CPI – Food

- **Source & Processing:**
  Obtained from BLS CPI dataset. Standardized during preprocessing.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  Identified as the most significant negative predictor of consumer sentiment across all models

4. CPI – Shelter

- **Source & Processing:**
  BLS CPI dataset. Standardized and retained after feature selection.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  Initially hypothesized as primary driver but found less impactful than food

5. CPI – Transportation

- **Source & Processing:**
  BLS CPI dataset. Included after removing multicollinear variables.
- **Observations: ~554**
- **Data Type: Quantitative**
- **Role: Independent variable**
- **Notes:**
  Exhibits higher volatility compared to shelter

6. CPI – Food at Home / Food Away from Home (Excluded Variables)

- **Source & Processing:**
  Subcomponents of CPI Food category
- **Reason for Exclusion:**
  - **Highly correlated with aggregate Food CPI**
  - **Redundant in regression model**

7. Standardized CPI Variables

- **Processing Step:**
  All CPI variables were standardized (z-scores) before modeling
- **Purpose:**
  - **Ensures comparability across variables**
  - **Prevents scale bias in regression coefficients**

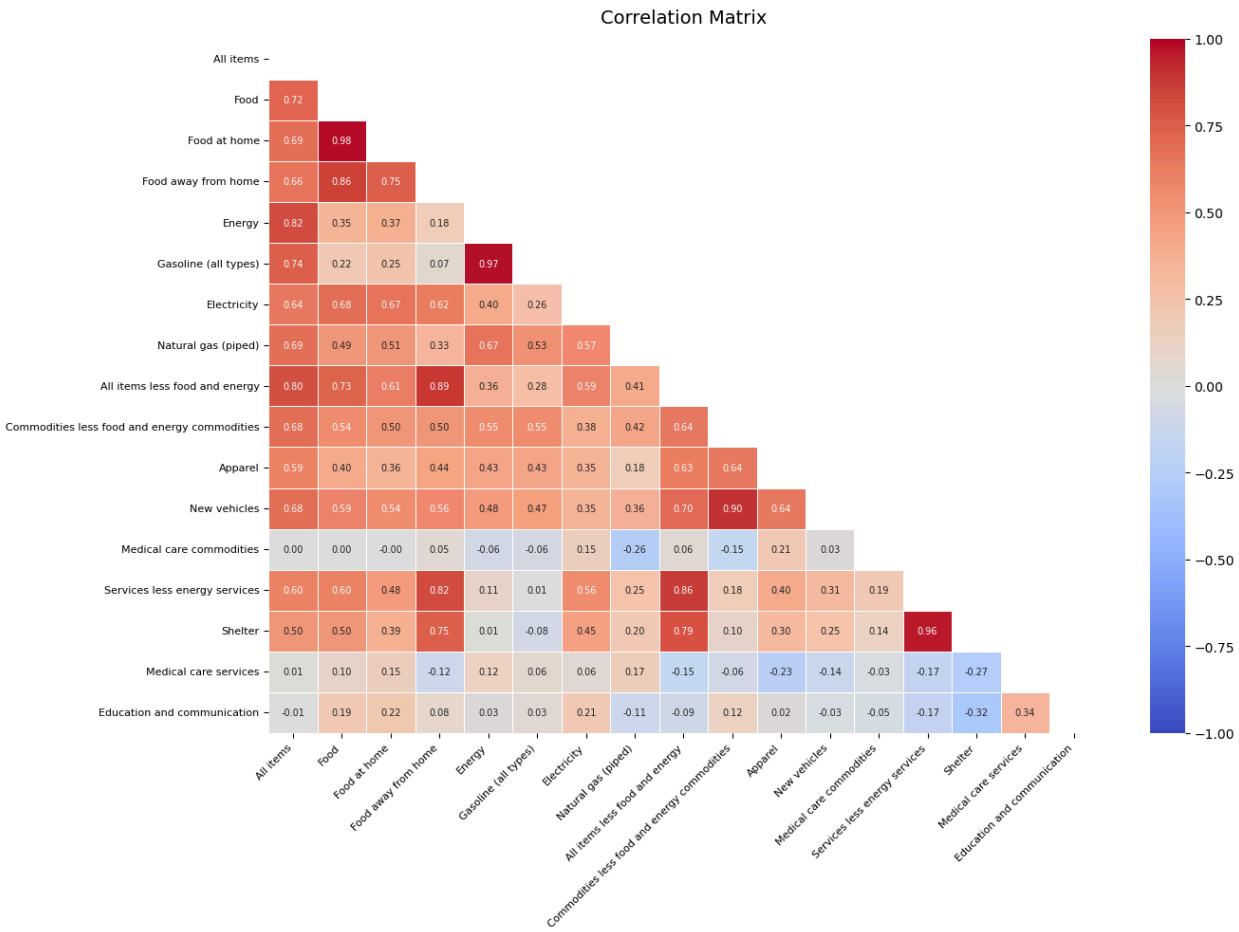
8. COVID Structural Break Indicator (Derived Variable)

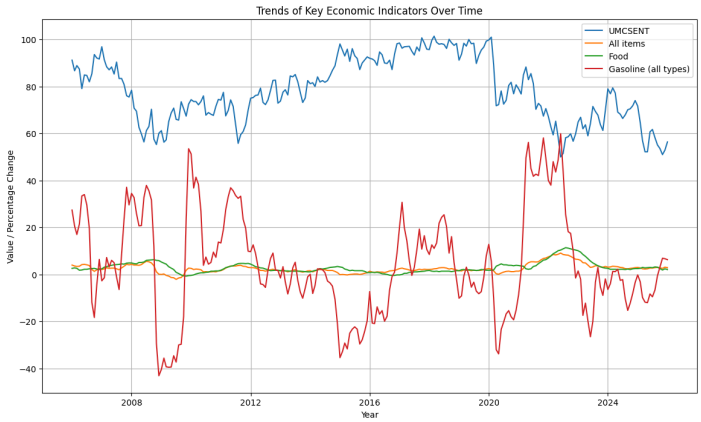
- **Source & Processing:**
  Created as a binary variable:
  - **0 = Pre-2020**
  - **1 = Post-2020**
- **Data Type: Binary**
- **Purpose:**
  Captures structural break identified through ADF testing and model failure post-2020

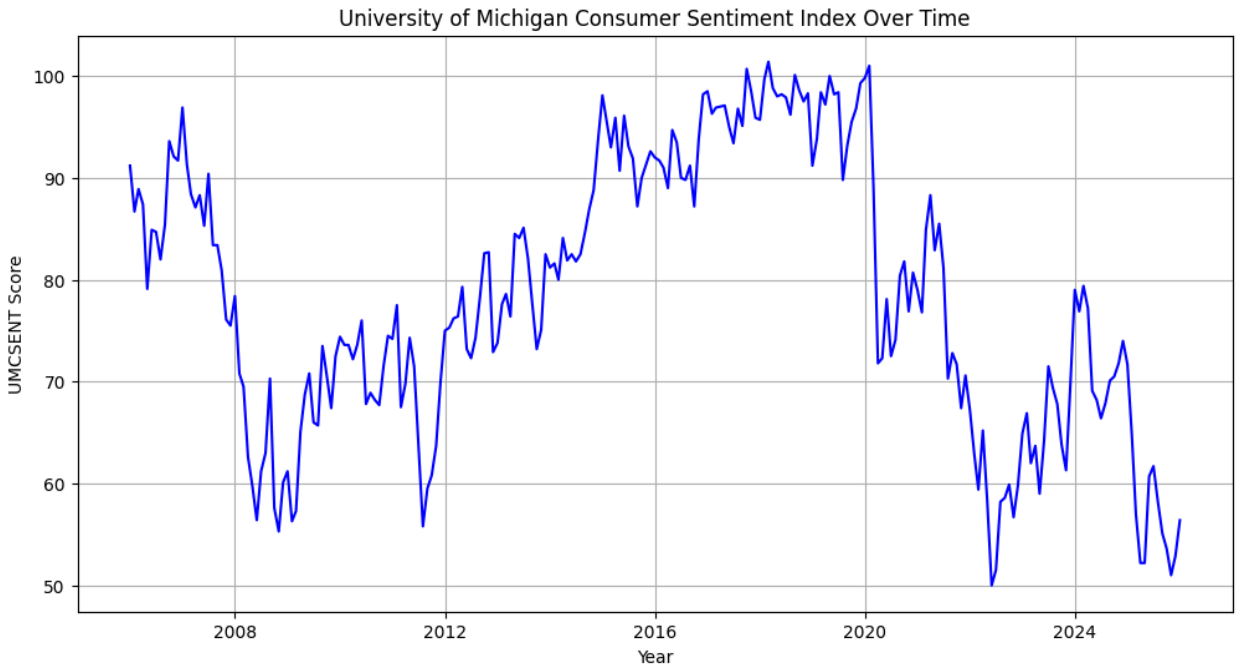
Data Cleaning & Preprocessing Summary

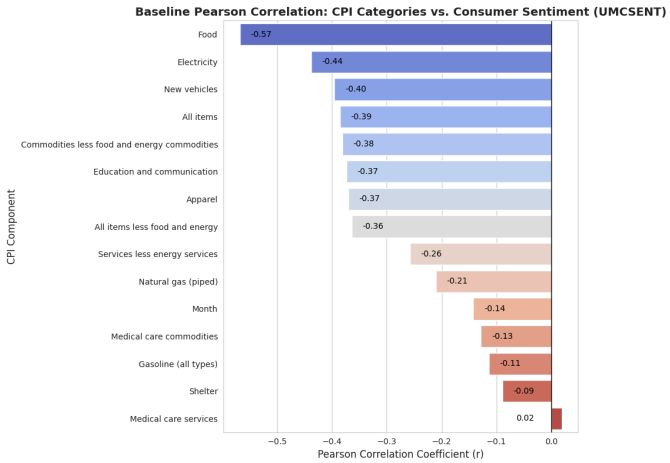
- **Merged CPI and UMCSENT datasets using monthly timestamps**
- **Restricted analysis window (for some models): 2006–2025**
- **Removed missing values (e.g., October 2025)**
- **Standardized CPI features**
- **Removed multicollinear variables using correlation threshold ($|r| \geq 0.85$)**
- **Identified a structural break around 2020 (COVID impact)**

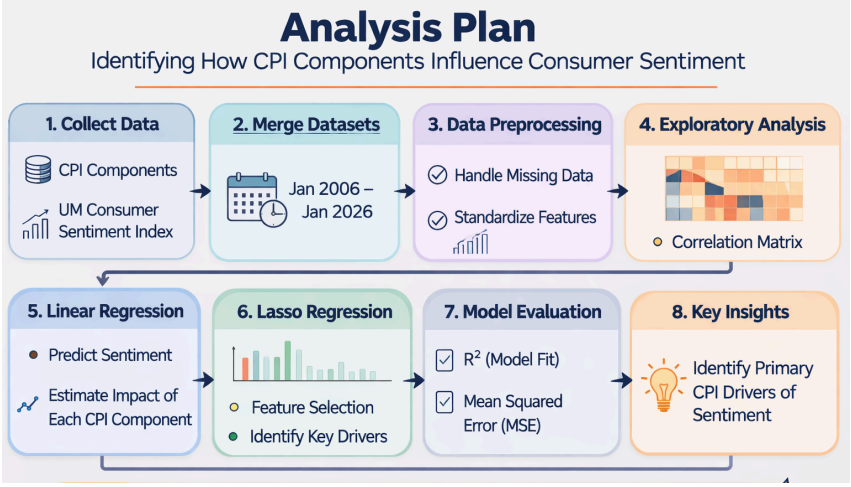
Visualization:


Correlation Matrix
All items
Food
Food at home
Food away from home
Energy
Gasoline (all types)
Electricity
Natural gas (piped)
All items less food and energy
Commodities less food and energy commodities
Apparel
New vehicles
Medical care commodities
Services less energy services
Shelter
Medical care services
Education and communication
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Trends of Key Economic Indicators Over Time
UMCSENT
All items
Food
Gasoline (all types)
Value / Percentage Change
Year
100
80
60
40
20
0
−20
−40
2008
2012
2016
2020
2024


University of Michigan Consumer Sentiment Index Over Time
UMCSENT Score
100
90
80
70
60
50
2008
2012
2016
2020
2024
Year


Baseline Pearson Correlation: CPI Categories vs. Consumer Sentiment (UMCSENT)
CPI Component
Food -0.57
Electricity -0.44
New vehicles -0.40
All items -0.39
Commodities less food and energy commodities -0.38
Education and communication -0.37
Apparel -0.37
All items less food and energy -0.36
Services less energy services -0.26
Natural gas (piped) -0.21
Month -0.14
Medical care commodities -0.13
Gasoline (all types) -0.11
Shelter -0.09
Medical care services 0.02
−0.5
−0.4
−0.3
−0.2
−0.1
0.0
Pearson Correlation Coefficient (r)


Analysis Plan
Identifying How CPI Components Influence Consumer Sentiment
1. Collect Data
CPI Components
UM Consumer Sentiment Index
2. Merge Datasets
Jan 2006 – Jan 2026
3. Data Preprocessing
Handle Missing Data
Standardize Features
4. Exploratory Analysis
Correlation Matrix
5. Linear Regression
Predict Sentiment
Estimate Impact of Each CPI Component
6. Lasso Regression
Feature Selection
Identify Key Drivers
7. Model Evaluation
R² (Model Fit)
Mean Squared Error (MSE)
8. Key Insights
Identify Primary CPI Drivers of Sentiment